

Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar

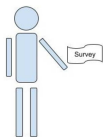
September 2026

The basic problem

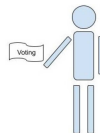
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

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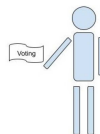
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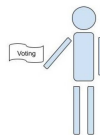
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We will be motivated in part by “Multilevel regression and poststratification” (MrP) [Gel97]:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[y|x_j]$.

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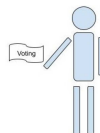
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

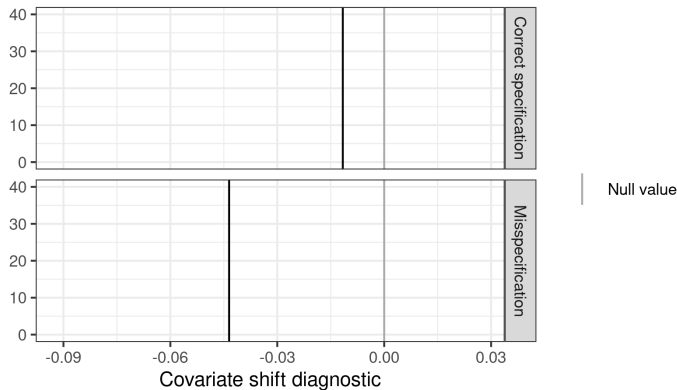


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

Result preview

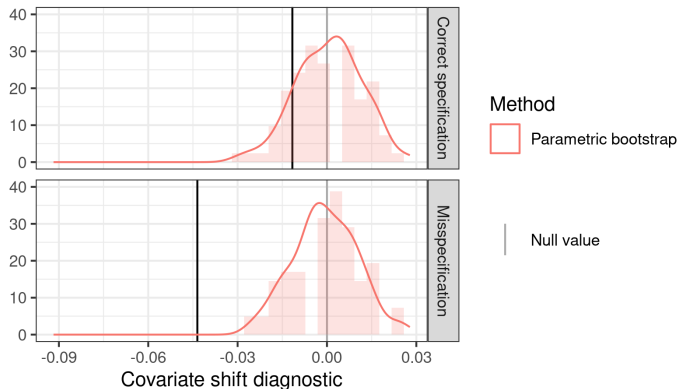


Figure 1: A covariate shift diagnostic

The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [[GB26](#)]

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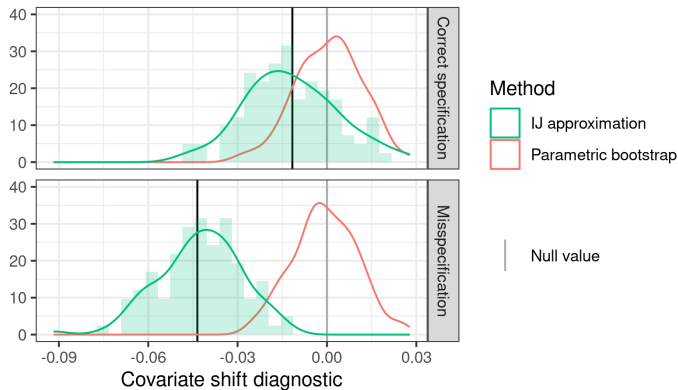


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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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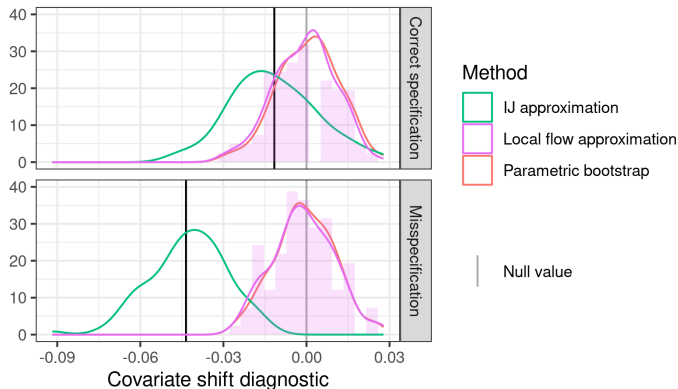
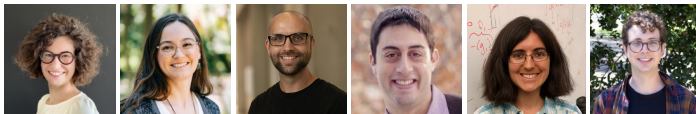


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Promising answer: Local flow sensitivity measures produce good approximations to the bootstrap at negligible computational expense.

- Local checks of specification of Bayesian regression models
 - (With Alice Cima, Lucas Schwengber, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Bayesian robustness with local flows
 - (With Lucas Schwengber)



Local checks of specification of Bayesian regression models

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \underbrace{\left[\overbrace{\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)}^{\text{Just a function } \phi(\theta)} \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Use out-of-sample checking [VGG17; Gel+20; Kuh+24]
 - The task is not to predict on a new IID survey datapoint, but to predict in a new population
- Specify an alternative model, maybe test with Bayes factor [DRM25; CBR25]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account

Local specification checks

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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

We will check the “stability” component in the PCS framework [YB24].

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This is what we actually want — let's check this directly.

Standard idea in causal, kernel learning, transfer learning [Shi00; Böh20; Roj+; MJK20].

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We provide a local Bayesian diagnostic version of this idea:

1. Let $\mathcal{P}_{\mathcal{D}}(x, y)$ denote a generic joint density of x, y .
2. Write our estimand as a functional $\hat{\mu}(\mathcal{P}_{\mathcal{D}})$
3. Define an additive covariate density perturbation $w(x)$
4. Define $\mathcal{P}_{\mathcal{D}}^\epsilon(x, y) = (\mathcal{P}_{\mathcal{D}}(x) + \epsilon w(x)) \mathcal{P}_{\mathcal{D}}(y|x)$
5. Compute the derivative $\Delta(\mathcal{D}) = \partial \hat{\mu}(\mathcal{P}_{\mathcal{D}}^\epsilon) / \partial \epsilon$ and check whether ≈ 0

Simple visual example

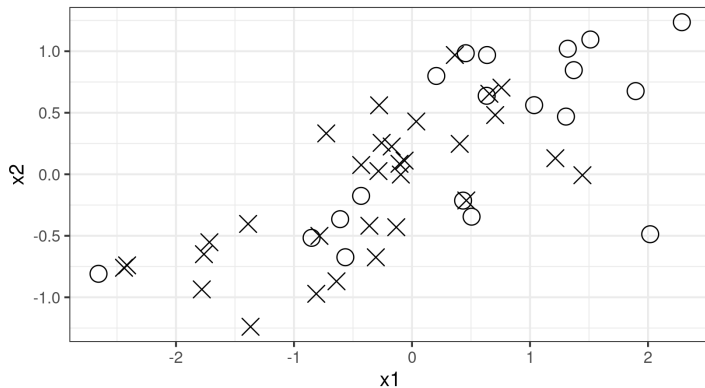


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

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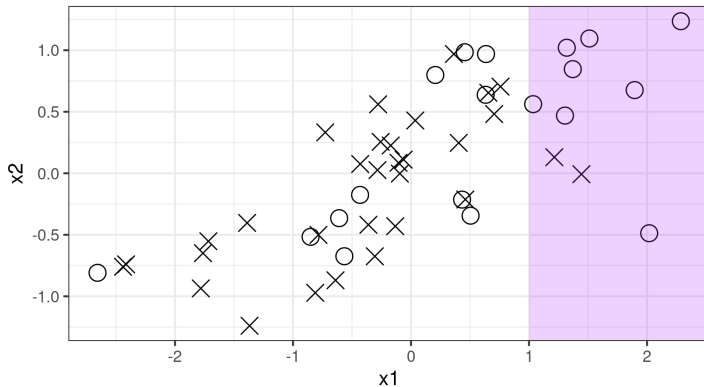


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Taking $w(x) = \mathbb{I}(x_1 > 1)$, we tend to upweight more positive examples.

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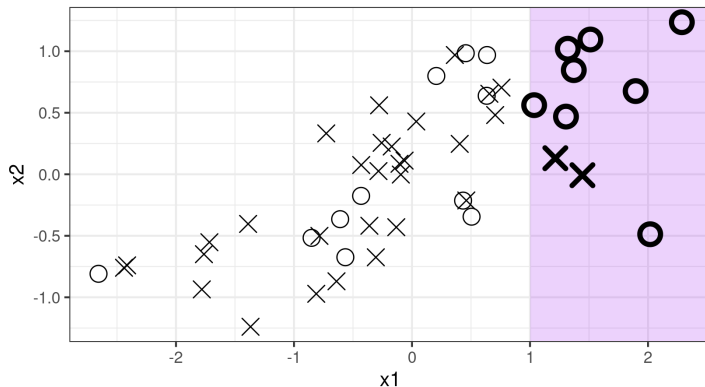


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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

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Theorem: [DF86; Gus96; [GBJ18](#)]:

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a function $\phi(\theta)$.

By interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left. \frac{\partial \hat{\mu}}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\underbrace{\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta)}_{\text{Log model perturbation}} \right)$$

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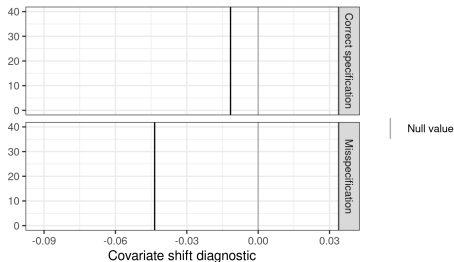


Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Correctly specified model: $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

Misspecified model: $y \sim \text{Logistic}(1 + X3 + X4 + X5)$

Diagnostic uses $w(x) = \mathbb{I}(X2 > 0.878)$

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To assess “closeness to zero,” compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \dots, 100$:
 1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \dots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ (**time consuming!**)
 3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare $\Delta(\mathcal{D})$ with distribution of $\Delta(\mathcal{D}_b^*)$.

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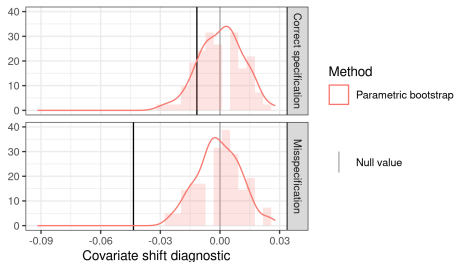


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Frequentist variance with the infinitesimal
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Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

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No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability meaningful under misspecification [HM23; WDH26; LJ26; JLR25]

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

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If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Task: Prove that the series is asymptotically accurate enough for hypothesis testing.

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, we study the *von Mises expansion* [von47].

Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t} \right|_{t=0} (1-0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0, 1]) \\ &= \underbrace{\sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem: [GB26] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary: [von47] Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon (\ell(y_i^*|x_i, \theta) - \ell(y_i|x_i, \theta)) \right).$$

(See, e.g., [Giordano et al. \[Gio+26\]](#) Theorem 4.1).

Bayesian von-Mises Expansion Results

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Note: The von-Mises expansion is stronger than necessary for consistency [GB26]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

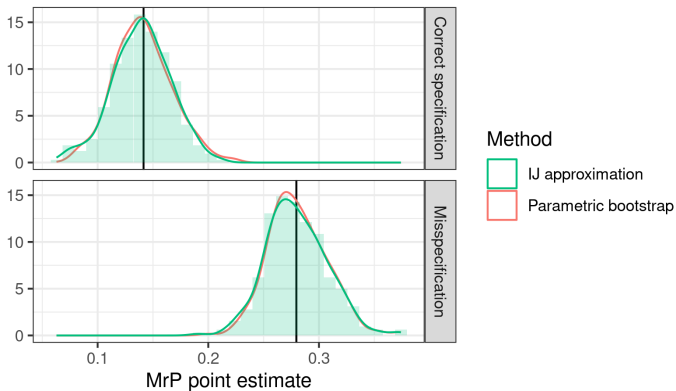


Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate.

See also Luo and Ji [LJ26] and Ji, Lee, and Rabe-Hesketh [JLR25].

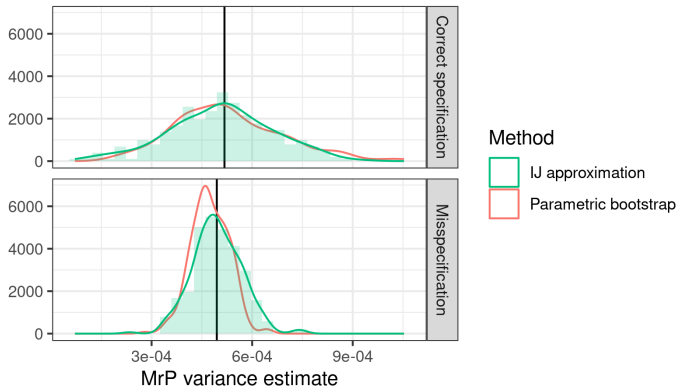


Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

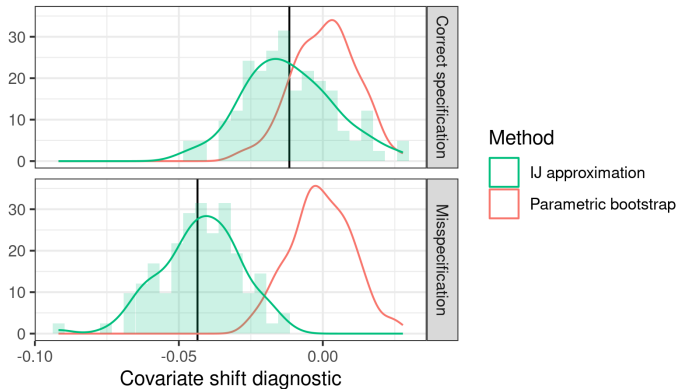


Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Bayesian robustness with local flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] \approx \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated with MCMC [Gus96; [GBJ18](#); CBR25; DRM25; [Gio+26](#)].

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This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations [HGV25]

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Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^\epsilon$.

There's a large literature on learning flows between distant distributions. [Pap+21]

Our problem is easier than much of this literature:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

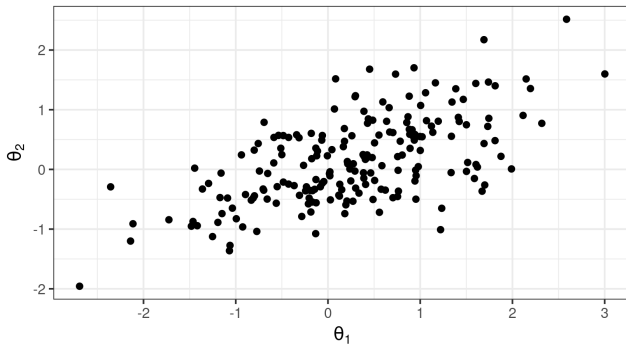


Figure 5: Draws from a posterior

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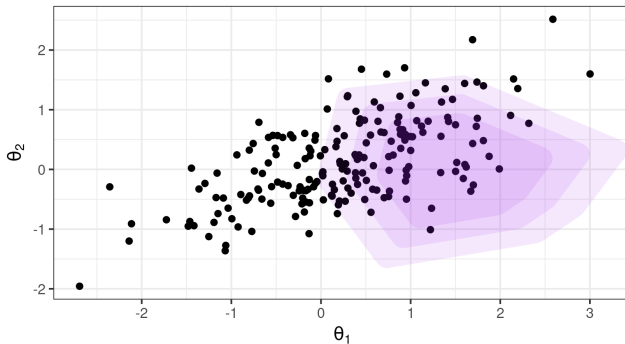


Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined “vertically.”

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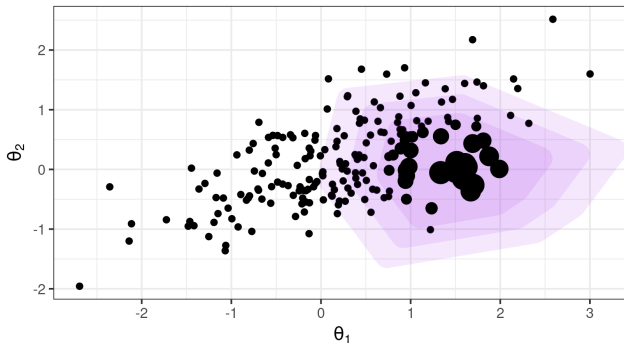


Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting [\[GBJ18, Appendix B\]](#)

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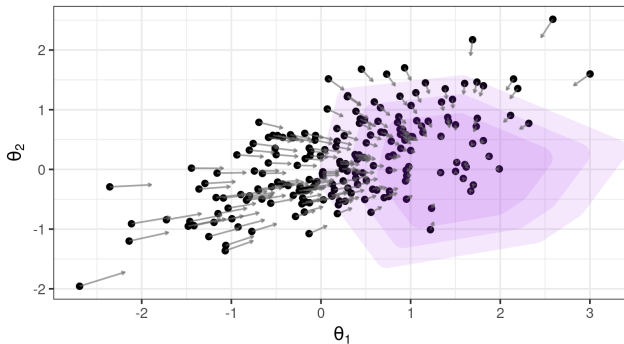


Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto G^\epsilon(\theta)$

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We want a flow $\theta \mapsto G^\epsilon(\theta)$ such that, for small ϵ , $G^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $G^0(\theta) = \theta \sim \mathcal{P}(\theta)$.

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Theorem: [KV86; BGL14]: Assume that $\mathbb{E}_{\mathcal{P}(\theta)} [h(\theta)^2] < \infty$ and $\mathcal{P}(\theta)$ satisfies a Poincare inequality. Then there exists a solution u to the partial differential equation

$$\mathcal{L}_{\mathcal{P}}(u) := -\nabla u^\top \nabla \log \mathcal{P} - \Delta u = h$$

such that $\|\nabla u\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon \nabla u(\theta)$ satisfies Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of a Langevin diffusion, and a Stein operator [ABS+21; Son+21; Ros11].

How can we find $u(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $u(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla u(\theta)] \quad (\text{Identity connecting } h \text{ and } u)$$

Basis expansion approach (the Petrov-Galerkin method [EG04; Nüs24; MM24]):

- Specify a family of real-valued target functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $u(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

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Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \text{Cov}_{\mathcal{P}(\theta)}(\phi_n(\theta), h(\theta)) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} = \underbrace{\begin{bmatrix} \vdots & & \\ \dots & \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi_n(\theta)^\top \nabla b_m(\theta)] & \dots \\ \vdots & & \end{bmatrix}}_{N \times M \text{ matrix}} \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

Theorem [GK69; BGL14], Schwengber 2026: When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $\mathcal{L}_{\mathcal{P}}^{-1}$ in its operator norm.

Using the 1^{st} order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_n$, the estimated flow is given by

$$\theta_{\epsilon} = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

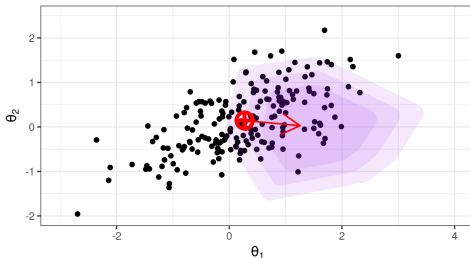
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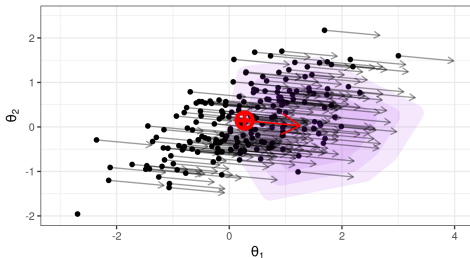
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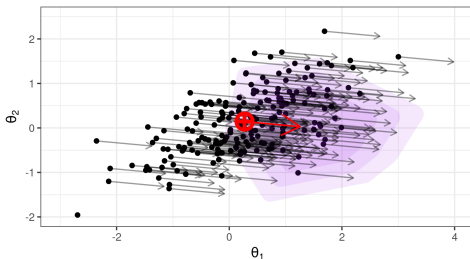
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Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i .

Set $\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

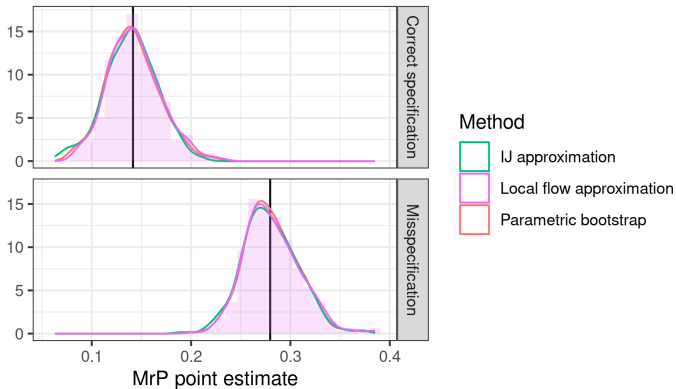


Figure 6: MrP Posterior Mean

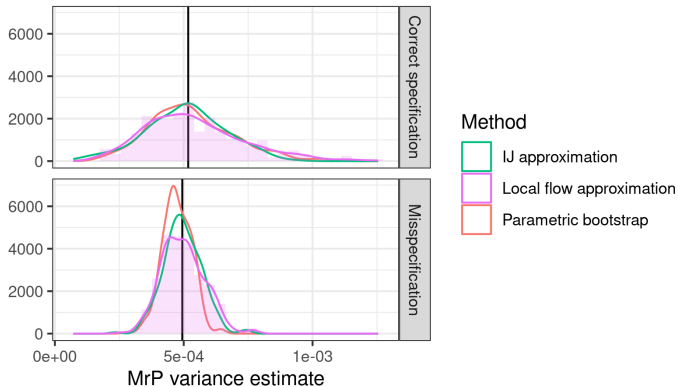


Figure 6: MrP Posterior Variance

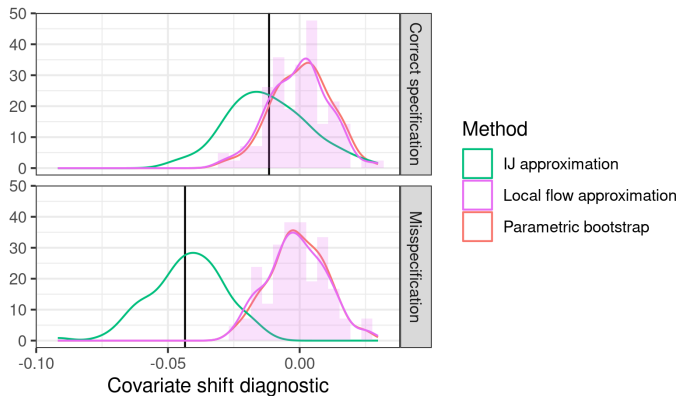


Figure 6: Covariate Shift Diagnostic

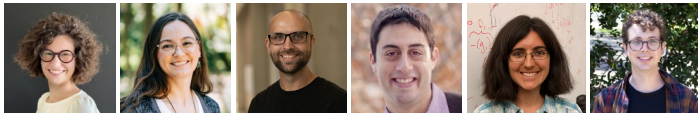
Carefully understanding the gap between IJ and the flow is ongoing work!

- Local checks of specification of Bayesian regression models
 - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Bayesian robustness with local flows
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Conclusion

- Local checks of specification of Bayesian regression models
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Lots of exciting work to to do!



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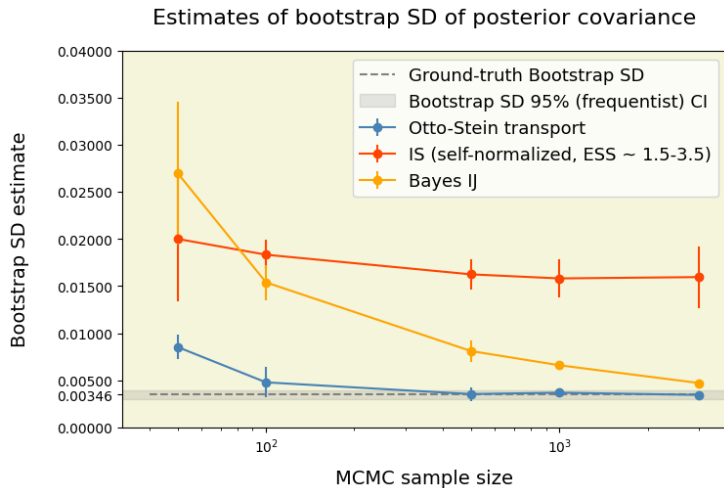
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Data re-weighting.

Augment the problem with *data weights* $w_1, \dots, w_N$. We can write $\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)]$.

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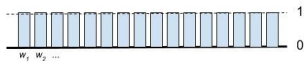


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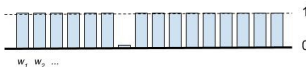
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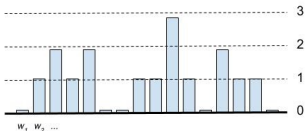
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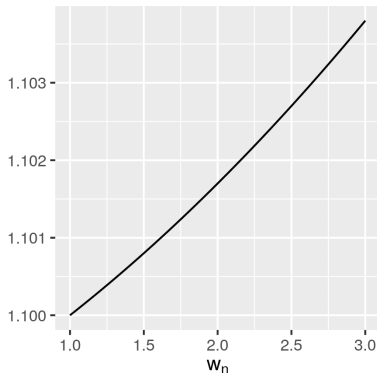
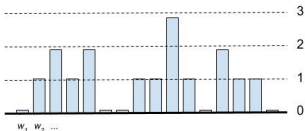
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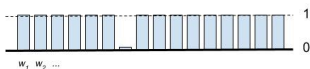
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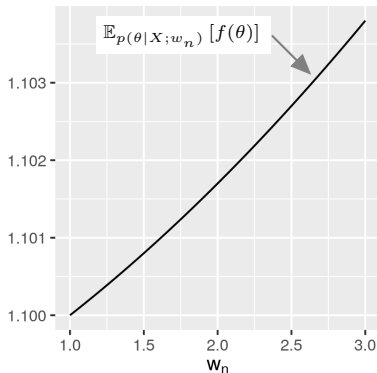
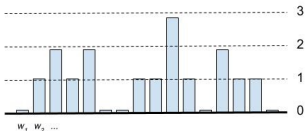
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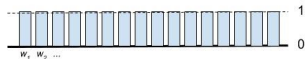
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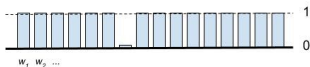
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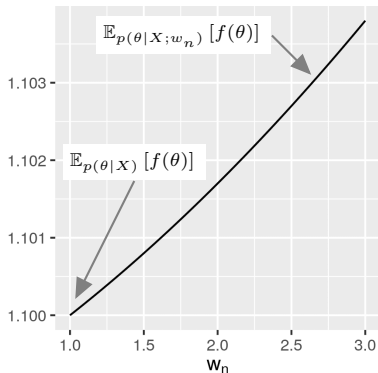
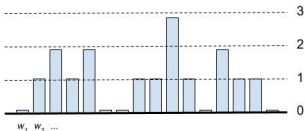
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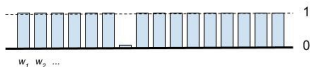
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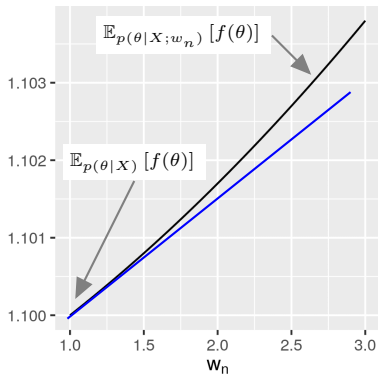
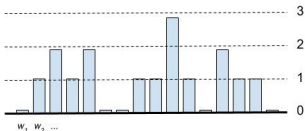
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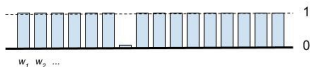
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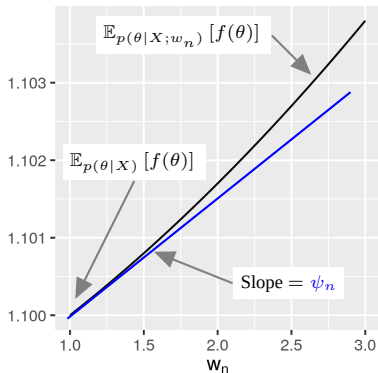
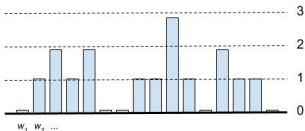
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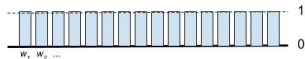
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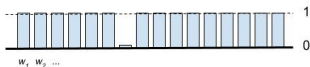
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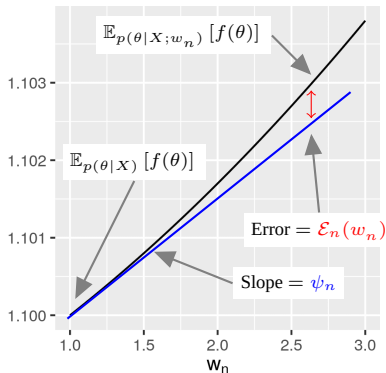
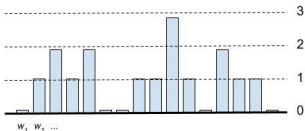
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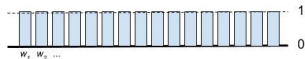
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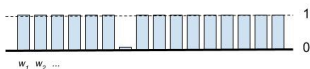
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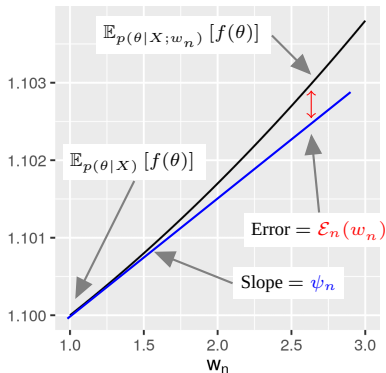
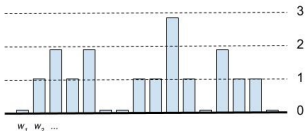
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The re-scaled slope $N\psi_n$ is known as the “influence function” at data point x_n .

$$\mathbb{E}_{\mathcal{P}(\theta|X;w)} [f(\theta)] - \mathbb{E}_{\mathcal{P}(\theta|X)} [f(\theta)] = \sum_{n=1}^N \psi_n (w_n - 1) + \mathcal{E}_n(w)$$

How can we use the approximation?

Assume the **slope** is computable and **error** is small.

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Bootstrap. Draw bootstrap weights $w \sim \mathcal{P}(w) = \text{Multinomial}(N, N^{-1})$.

$$\text{Bootstrap variance} = \text{Var}_{\mathcal{P}(w)} \left(\mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right)$$

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$$\begin{aligned} \text{Bootstrap variance} &= \text{Var}_{\mathcal{P}(w)} \left(\mathbb{E}_{p(\theta|X;w)} [f(\theta)] \right) \\ &= \text{Var}_{\mathcal{P}(w)} \left(\sum_{n=1}^N \psi_n(w_n - 1) + \mathcal{E}_n(w) \right) \\ &= \frac{1}{N^2} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 + \text{Term involving } \mathcal{E}_n(w) \text{ for } n = 1, \dots, N \\ &\approx \frac{1}{N} \underbrace{\left(\frac{1}{N} \sum_{n=1}^N \left(\psi_n - \bar{\psi} \right)^2 \right)}_{\text{"Infinitesimal jackknife variance estimate"}} \end{aligned}$$

Bayesian posterior functionals

Suppose we are interested in $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)]$. Let the regression log likelihood be $\ell(y|x, \theta)$.

Define the “generalized posterior” functional [GB26]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Let $\hat{\mathcal{P}}_S$ denote the empirical distribution over x_i, y_i . Then

$$T(\hat{\mathcal{P}}_S, N_S) = \frac{\int g(\theta) \exp \left(N_S \frac{1}{N_S} \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) d\theta}{\int \exp \left(N_S \frac{1}{N_S} \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \right) \pi(\theta) d\theta} = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [g(\theta)] .$$

Consider a smooth path of densities $\epsilon \mapsto \mathcal{P}^\epsilon$. The directional derivative in general is

$$\begin{aligned} \left. \frac{\partial T(\mathcal{P}^\epsilon, N)}{\partial \epsilon} \right|_{\epsilon=0} &= \text{Cov}_{\mathcal{P}(\theta|\mathcal{P}, N)} \left(g(\theta), N \int \ell(y|x, \theta) w(dx) \left. \frac{\partial \log \mathcal{P}_\epsilon(y|x)}{\partial \epsilon} \right|_{\epsilon=0} dx dy \right) \\ \Rightarrow \left. \frac{\partial T(\hat{\mathcal{P}}_S^\epsilon, N_S)}{\partial \epsilon} \right|_{\epsilon=0} &= \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(g(\theta), \sum_{i=1}^{N_S} \ell(y_i|x_i, \theta) \left. \frac{\partial \log \hat{\mathcal{P}}_S^\epsilon(y_i|x_i)}{\partial \epsilon} \right|_{\epsilon=0} \right) \end{aligned}$$

Bayesian posterior functionals

We're interested in $g(\theta) = \frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)$.

For some function $w(x)$, set

$$\mathcal{P}_S^\epsilon(x, y) = \mathcal{P}_S(x, y) + \epsilon w(dx) \mathcal{P}_S(x, y) = (\mathcal{P}_S(x) + \epsilon w(x)) \mathcal{P}_S(y|x).$$

This is a shift in the marginal distribution of x in the direction $w(x)$, with

$$\left. \frac{\partial \mathcal{P}_S^\epsilon(y|x)}{\partial \epsilon} \right|_{\epsilon=0} = w(x) \mathcal{P}_S(y|x).$$

Our diagnostic is then given by

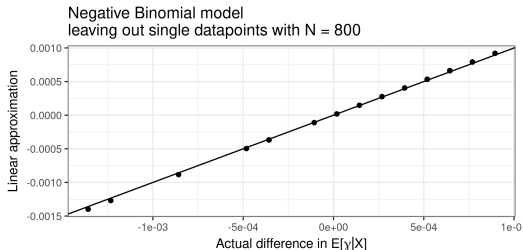
$$\Delta(\mathcal{P}) = \text{Cov}_{\mathcal{P}(\theta)} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j), N \int \ell(y|x, \theta) \mathcal{P}_S(y|x) w(x) dx dy \right)$$

Exchangeable units: Experimental results revisited

Our results were actually computed on **identical datasets** with $G = N$ and $g_n = n$.

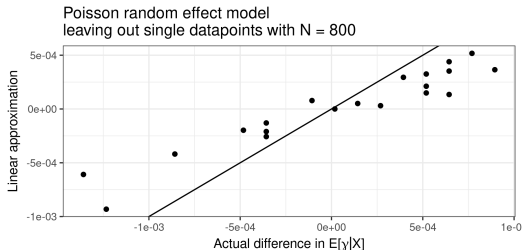
Uses $\log \mathcal{P}(x_n|\gamma)$:

$$\psi_n = \mathbb{E}_{\mathcal{P}(\gamma|X)} [\bar{\gamma} \bar{\ell}_n(\gamma)]$$



Uses $\log \mathcal{P}(x_n|\gamma, \lambda)$:

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Not easily computable from

$$\gamma, \lambda \sim \mathcal{P}(\gamma, \lambda|X)$$

in general.

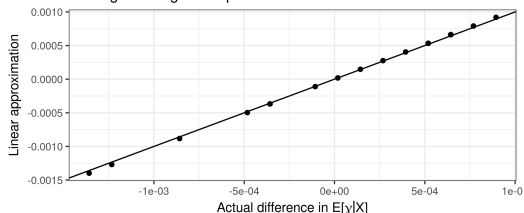
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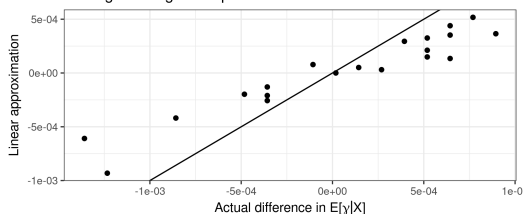
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Negative Binomial model
leaving out single datapoints with $N = 800$



Poisson random effect model
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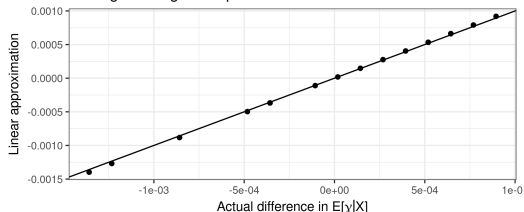
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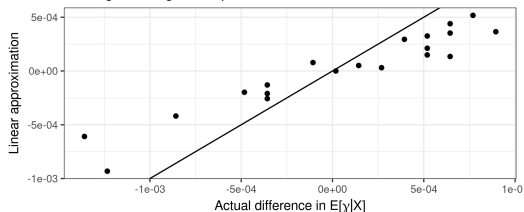
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May still be useful when $\mathcal{P}(\lambda|X)$ is *somewhat* concentrated.

Negative Binomial model
leaving out single datapoints with $N = 800$



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Example: Exponential families with random effects (REs) λ and fixed effects γ .

If the observations per random effect remains bounded as $N \rightarrow \infty$, then

- Parameter λ (“local”) grows in dimension with N .
- Parameter γ (“global”) is finite-dimensional.
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In general, we cannot hope for an asymptotic analysis of $\mathbb{E}_{\mathcal{P}(\lambda, \gamma|X)} [f(\lambda)]$.

Can we save the approximation when *some* parameters concentrate?

Does the residual vanish asymptotically for $w_n \mapsto \mathbb{E}_{\mathcal{P}(\gamma|X; w_n)} [f(\gamma)]$?

We assume that $\mathcal{P}(\gamma|X)$ concentrates but $\mathcal{P}(\lambda|X)$ does not. By our series expansion:

$$\mathbb{E}_{\mathcal{P}(\gamma, \lambda|X; w_n)} [\gamma] - \mathbb{E}_{\mathcal{P}(\gamma, \lambda|X)} [\gamma] =$$
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Corollary [GB26]:

In general, $w_n \mapsto N \left(\mathbb{E}_{\mathcal{P}(\gamma|X; w_n)} [\gamma] - \mathbb{E}_{\mathcal{P}(\gamma|X)} [\gamma] \right)$ remains non-linear as $N \rightarrow \infty$.

Example: Poisson regression with Gamma-distributed random effects

For $g = 1, \dots, G$, $\lambda_g \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$ for fixed α, β

For $n = 1, \dots, N$, $g_n \stackrel{iid}{\sim} \text{Categorical}(1, \dots, G)$, $y_n | \lambda_n, \gamma, g_n \stackrel{iid}{\sim} \text{Poisson}(\gamma \lambda_{g_n})$.

$x_n = (y_n, g_n)$ are IID given λ, γ . Write $\log \mathcal{P}(X | \lambda, \gamma; w) = \sum_{n=1}^N w_n \ell_n(\lambda, \gamma)$.

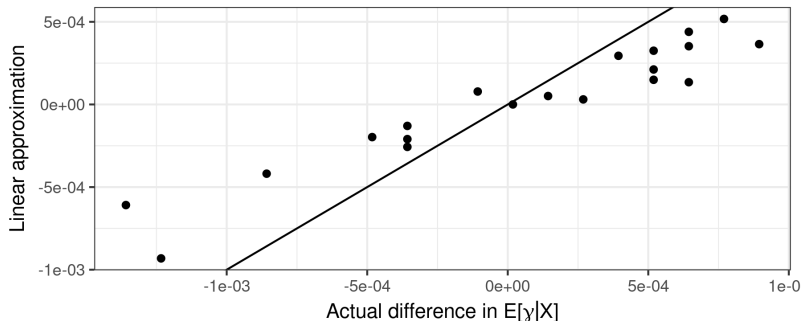
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Poisson random effect model
leaving out single datapoints with $N = 800$



Theorem 3 [GB26] (sketch):

(Consistency of the von-Mises expansion in finite dimensions)

Under slightly stronger conditions our original finite-dimensional posterior consistency result,

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \rightarrow 0 \quad \text{in the Bayesian von-Mises expansion.}$$

Bayesian von-Mises Expansion Results

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Theorem 4 [GB26] (sketch):

(Inconsistency of the von-Mises expansion in high dimensional exponential families)

Assume that x_n comes with a equiprobable group assignment $g_n \in 1, \dots, G$.

Conditional on g , x_n is modeled as a finite-dimensional exponential family given λ, γ :

$$\log \mathcal{P}(x_n | g_n = g, \gamma, \lambda) = \tau(x_n)^\top \eta_g(\gamma, \lambda) + \text{Constant.}$$

Define the average product of second moments:

$$\mathcal{V}_N(\gamma) := \frac{1}{G} \sum_{g=1}^G \text{tr} \left(\mathbb{E}_{\mathbb{F}(x_n)} [\tau(x_n) \tau(x_n)^\top] \text{Cov}_{\mathcal{P}(\lambda|\gamma, \mathbb{F})} (\eta_g(\gamma, \lambda)) \right).$$

If $N \mathbb{E}_{\mathcal{P}(\gamma|\mathbb{F})} [\bar{f}(\gamma) \mathcal{V}_N(\gamma)]$ is strictly bounded away from 0 as $N \rightarrow \infty$, then

$$\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \rightarrow \infty \quad \text{in the Bayesian von-Mises expansion.}$$